AN ETALE COVER THAT CANNOT BE REFINED BY A FINITE COMPOSITION OF OPEN IMMERSIONS AND FINITE ETALE MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Context span 4444-4452 included per the situation-above precedent; without it the statement's 'the etale cover U -> X' dangles.

Editorial note. The label lemma-etale-not-dominated-zariski-finite-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The section construction carried as context span: X a smooth connected affine curve over C, g : Y -> X finite of degree 3 with fibre over x equal to an unramified point u (degree 1) and a ramified point y (degree 2), U = Y \ {y} and f = g|_U : U -> X etale surjective with f^{-1}(x) = {u}
- Discrete valuation rings, ramification indices, connected components of smooth curves; fibres of finite morphisms

Let X be a smooth, connected affine curve over $\mathbf{C}$. Choose a finite morphism $g : Y \to X$ of degree 3 from a smooth connected curve Y such that there is a point $x \in X$ whose fiber $g^{-1}(x) = \{u, y\}$ consists of an unramified point u (degree 1) and a ramified point y (degree 2). By replacing X with a Zariski open neighborhood of x, we may assume g is unramified everywhere except at y. Let $U = Y \setminus \{y\}$. Then $f = g|_U : U \to X$ is an étale surjective morphism, and $f^{-1}(x) = \{u\}$.

Lemma 0.1. *The étale cover $U \to X$ cannot be refined by any finite composition of open immersions and finite étale morphisms.*

Proof. Towards contradiction, assume there exists a tower of morphisms

$$W_m \to W_{m-1} \to \cdots \to W_1 \to W_0 = X$$

where each map $W_i \to W_{i-1}$ is either an open immersion or a finite étale morphism, a lift $W_m \to U$ of $W_m \to X$ along $f : U \to X$, and a point $w \in W_m$ mapping to $u \in U$ (and thus to $x \in X$). By replacing each W_k with the connected component containing the image $w_k \in W_k$ of w, we may assume that each W_k is smooth and connected.

For each $0 \leq i \leq m$, consider the fiber product $Z_i = Y \times_X W_i$. As $W_i \to X$ is étale and Y is smooth, the curve Z_i is a disjoint union of smooth, connected curves and the maps $Z_i \to W_i$ are finite of degree 3 by base change. Clearly, the fibre of

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$Z_i \to W_i$ over w_i consists of two points z_i, v_i mapping to y, u in Y. The morphism $Z_i \to W_i$ is unramified at v_i and ramified of degree 2 at z_i. Let $n_i = \#\pi_0(Z_i)$. Since the fibre of $Z_i \to W_i$ has 2 points we see that $n_i \in \{1, 2\}$. Moreover, n_i can only increase with i as the transition maps $Z_i \to Z_{i-1}$ are dominant. We clearly have $n_0 = 1$. Moreover, we must have $n_m \geq 2$: there is an X-morphism $W_m \to U \subset Y$, so the base change $Z_m \to W_m$ has a section. There is then a minimal index $i \geq 1$ where $n_{i-1} = 1$ and $n_i = 2$. As the n_j's are determined by whether or not Z_i is irreducible, the map $W_i \to W_{i-1}$ cannot be an open immersion, so it must be finite étale. As $n_i = 2$, the smooth curve Z_i has two connected components; as the map to W_i is finite, one of these components must then map isomorphically to W_i, so we can write $Z_i = W_i \sqcup Z_i'$, where $Z_i' \to W_i$ has degree 2. The induced map $W_i \subset Z_i \to Z_{i-1}$ is a dominant map between connected curves, both of which are finite over W_{i-1}. Any such map must be surjective. Thus we may pick a point $t \in W_i$ mapping to z_{i-1}. We obtain extensions

$$\mathcal{O}_{W_{i-1}, w_{i-1}} \to \mathcal{O}_{Z_i, z_i} \to \mathcal{O}_{W_i, z}$$

of discrete valuation rings. Now the first one is ramified while the composite is unramified as $W_i \to W_{i-1}$ was étale; this is impossible, so we obtain a contradiction. $\square$